

Kai Zhao

Assistant Professor · Department of Computer Science · Florida State University

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RESEARCH INTERESTS

- High-Performance Computing
- Scientific Data Management
- Error-Bounded Data Reduction
- Trustworthy and Efficient AI

EDUCATION

Ph.D. in Computer Science	2017–2022
University of California, Riverside	Riverside, CA
B.S. in Computer Science	2010–2014
Peking University	Beijing, China

PROFESSIONAL EXPERIENCE

■ Assistant Professor, Florida State University, Tallahassee, FL	2023–present
■ Assistant Professor, University of Alabama at Birmingham, Birmingham, AL	2022–2023
■ Research Aide, Argonne National Laboratory, Lemont, IL	2019
■ Senior Software Engineer, Wacai Technology Co., Ltd., Hangzhou, China	2017
■ Software Engineer, MicroStrategy, Hangzhou, China	2016
■ Software Engineer, Alibaba Group, Hangzhou, China	2014–2016

HONORS AND AWARDS

■ CAREER Award, NSF	2026
■ Best Paper Nominee, ACM HPDC'25	2025
■ Best Paper Award, IEEE Transactions on Big Data (SZ3)	2024
■ First Year Assistant Professor Award, Florida State University	2024
■ Best Paper Nominee, ACM ICS'23	2023
■ R&D 100 Award (SZ compression framework)	2021
■ Laxmi N. Bhuyan Fellowship, University of California, Riverside	2021
■ Dissertation Year Program Fellowship, University of California, Riverside	2021
■ Dean's Distinguished Fellowship, University of California, Riverside	2017

RESEARCH FUNDING

Total external funding: \$1.26M. All amounts listed are my portion.

■ PI , NSF CAREER, <i>StreamZ: An Adaptive and Scalable Lossy Compression Framework for Extreme-Scale Scientific Data Streaming Pipelines</i> , \$572K	2026–2031
■ Site PI , Argonne National Laboratory, <i>AI-Assisted Lossy Compression Design</i> , \$75K	2026–2027
■ Site PI , Argonne National Laboratory, <i>Lossy Compression for Streaming Data</i> , \$77K	2024
■ PI , Florida State University FYAP, <i>HLC: A Homomorphic Lossy Compression Framework for Scientific Applications</i> , \$20K	2024
■ Site PI , NSF CSSI, <i>Collaborative Research: Frameworks: FZ: A Fine-Tunable Cyberinfrastructure Framework to Streamline Specialized Lossy Compression Development</i> , \$540K	2023–2027

RESEARCH SOFTWARE

- **SZ family of error-bounded lossy compressors** (<https://github.com/szcompressor>; 2021 R&D 100 Award). I am the lead developer and maintainer of **SZ3**, a state-of-the-art error-bounded compressor for scientific data, and I lead the compression-module-library thrust of **FZ**, an NSF CSSI Frameworks project that unifies state-of-the-art compression techniques into reusable modules for building compressors tailored to each application. **SZ is used across science and AI, in domains including cosmology, Earth system science, X-ray light sources, seismic**

imaging, molecular dynamics, and quantum circuit simulation, and is integrated into DOE's ADIOS2 I/O framework and is being integrated into GROMACS's next-generation trajectory format; it is part of the E4S software stack on US exascale systems, ships in the ClickHouse database, and is distributed via Spack, PyPI, Julia, Arch Linux, FreeBSD, and Rust crates.

REFEREED CONFERENCE PUBLICATIONS

61 peer-reviewed papers; 35 at CSRankings venues; 3 best-paper honors. Underlined: students I advise.

- C1 [SC'26] Jiefeng Zhou, Sheng Di, Yanfei Guo, Yuhao Guo, Lingqi Zhang, **Kai Zhao**, Rajeev Thakur, Franck Cappello, Jiajun Huang, "hZCCL2: Co-Designing Collective Communication with Vectorized Homomorphic Compression." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, Chicago, IL, USA, Nov 15–20, 2026.
- C2 [SC'26] Changqing Li, Sheng Di, **Kai Zhao**, Wenqian Dong, "LLM Agents Meet Lossy Compression: Benchmark, Demystify and Optimize across HPC Architectures." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, Chicago, IL, USA, Nov 15–20, 2026.
- C3 [VLDB'26] Ning Yan, Sheng Di, **Kai Zhao**, Lipeng Wan, "Eureka: Enabling Fine-Grained Access and Range Queries on Compressed Scientific Data via Data-Index Co-Compression." *Proceedings of the International Conference on Very Large Data Bases*, Boston, MA, USA, Aug 31–Sep 4, 2026.
- C4 [VLDB'26] Youyuan Liu, Longtao Zhang, Ruoyu Li, Bo Jiang, Taolue Yang, **Kai Zhao**, Sheng Di, Eduard Dragut, Sian Jin, "Error-bounded Point Cloud Compression Using Truncated Octahedron Quantization." *Proceedings of the International Conference on Very Large Data Bases*, Boston, MA, USA, Aug 31–Sep 4, 2026.
- C5 [HPDC'26] Zhuoxun Yang, Ruoyu Li, Amit Subrahmanya, Vishwas Rao, Sheng Di, Robert Underwood, Longtao Zhang, Jinyang Liu, Franck Cappello, **Kai Zhao**, "TZ: Achieving High-Ratio Scientific Data Compression on GPUs with Global Data Decomposition." *Proceedings of the ACM International Symposium on High-Performance Parallel and Distributed Computing*, Cleveland, OH, USA, July 13–16, 2026.
- C6 [HPDC'26] Longtao Zhang, Ruoyu Li, Zhuoxun Yang, Robert Underwood, Sheng Di, Daoce Wang, Jinyang Liu, Jiajun Huang, Franck Cappello, **Kai Zhao**, "OPAL: On-demand Progressive Accelerated Scientific Lossy Compression." *Proceedings of the ACM International Symposium on High-Performance Parallel and Distributed Computing*, Cleveland, OH, USA, July 13–16, 2026.
- C7 [HPDC'26] Sujata Sinha, Sheng Di, Vishwas Rao, Robert Underwood, David Lenz, Zizhe Jian, Zhuoxun Yang, **Kai Zhao**, Lingjia Liu, Franck Cappello, "Bridging Information Theory and Practice for Scientific Lossy Compression." *Proceedings of the ACM International Symposium on High-Performance Parallel and Distributed Computing*, Cleveland, OH, USA, July 13–16, 2026.
- C8 [ICS'26] Ruoyu Li, Yafan Huang, Longtao Zhang, Zhuoxun Yang, Sheng Di, Boyuan Zhang, Jiajun Huang, Jinyang Liu, Jiannan Tian, Guanpeng Li, Fengguang Song, Hanqi Guo, Franck Cappello, **Kai Zhao**, "GPZ: GPU-Accelerated Lossy Compressor for Particle Data." *Proceedings of the ACM International Conference on Supercomputing*, Belfast, United Kingdom, July 6–9, 2026.
- C9 [ICDE'26] Xuan Wu, Sheng Di, Tripti Agarwal, **Kai Zhao**, Xin Liang, Franck Cappello, "Enabling Homomorphic Analytical Operations on Compressed Scientific Data with Multi-stage Decompression." *Proceedings of the IEEE International Conference on Data Engineering*, Montreal, QC, Canada, May 4–8, 2026.
- C10 [SC'25] Shixun Wu, Jinwen Pan, Jinyang Liu, Jiannan Tian, Ziwei Qiu, Jiajun Huang, **Kai Zhao**, Xin Liang, Sheng Di, Zizhong Chen, Franck Cappello, "Boosting Scientific Error-Bounded Lossy Compression through Optimized Synergistic Lossy-Lossless Orchestration." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, St. Louis, MO, USA, Nov 16–21, 2025.
- C11 [SC'25] Daoce Wang, Pascal Grosset, Jesus Pulido, Jiannan Tian, Tushar M. Athawale, Jinda Jia, Baixi Sun, Boyuan Zhang, Sian Jin, **Kai Zhao**, James Ahrens, Fengguang Song, "STZ: A High Quality and High Speed Streaming Lossy Compression Framework for Scientific Data." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, St. Louis, MO, USA, Nov 16–21, 2025.
- C12 [SC'25] Franck Cappello, Robert Underwood, Yuri Alexeev, Allison H. Baker, Ebru Bozdağ, Martin Burtscher, Kyle Chard, Sheng Di, Kyle Gerard Felker, Paul Christopher O'Grady, Hanqi Guo, Yafan Huang, Peng Jiang, Sian Jin, Petter Johansson, Shaomeng Li, Xin Liang, Erik Lindahl, Peter Lindstrom, Zarija Lukić, Magnus Lundborg, Danylo Lykov, Masaru Nagaso, Kento Sato, Amarjit Singh, Seung Woo Son, Shihui Song, William Tang, Dingwen Tao, Jiannan Tian, Kazutomo Yoshii, **Kai Zhao**, "What to Support When You're Compressing: The State of Practice, Gaps, and Opportunities for Scientific Data Compression." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, St. Louis, MO, USA, Nov 16–21, 2025.

- C13 [HPDC'25 (Best Paper Nominee)]** Zhuoxun Yang, Sheng Di, Longtao Zhang, Ruoyu Li, Ximiao Li, Jiajun Huang, Jinyang Liu, Franck Cappello, **Kai Zhao**, "IPComp: Interpolation Based Progressive Lossy Compression for Scientific Applications." *Proceedings of the ACM International Symposium on High-Performance Parallel and Distributed Computing*, Notre Dame, IN, USA, July 20–23, 2025.
- C14 [VLDB'25]** Jinyang Liu, Pu Jiao, **Kai Zhao**, Xin Liang, Sheng Di, Franck Cappello, "QPET: A Versatile and Portable Quantity-of-Interest-preservation Framework for Error-Bounded Lossy Compression." *Proceedings of the International Conference on Very Large Data Bases*, London, UK, Sep 1–5, 2025.
- C15 [SIGMOD'25]** Longtao Zhang, Ruoyu Li, Congrong Ren, Sheng Di, Jinyang Liu, Jiajun Huang, Robert Underwood, Pascal Grosset, Dingwen Tao, Xin Liang, Hanqi Guo, Franck Cappello, **Kai Zhao**, "LCP: Enhancing Scientific Data Management with Lossy Compression for Particles." *Proceedings of the ACM SIGMOD International Conference on Management of Data*, Berlin, Germany, June 22–27, 2025.
- C16 [SC'24]** Daoce Wang, Pascal Grosset, Jesus Pulido, Tushar M. Athawale, Jiannan Tian, **Kai Zhao**, Zarija Lukić, Axel Huebl, Zhe Wang, James Ahrens, Dingwen Tao, "A High-Quality Workflow for Multi-Resolution Scientific Data Reduction and Visualization." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, Atlanta, GA, USA, Nov 17–22, 2024.
- C17 [SC'24]** Jiajun Huang, Sheng Di, Xiaodong Yu, Yujia Zhai, Jinyang Liu, Zizhe Jian, Xin Liang, **Kai Zhao**, Xiaoyi Lu, Zizhong Chen, Franck Cappello, Yanfei Guo, Rajeev Thakur, "hZCCL: Accelerating Collective Communication with Co-Designed Homomorphic Compression." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, Atlanta, GA, USA, Nov 17–22, 2024.
- C18 [SC'24]** Jinyang Liu, Jiannan Tian, Shixun Wu, Sheng Di, Boyuan Zhang, Robert Underwood, Yafan Huang, Jiajun Huang, **Kai Zhao**, Guanpeng Li, Dingwen Tao, Zizhong Chen, Franck Cappello, "cuSZ-i: High-Ratio Scientific Lossy Compression on GPUs with Optimized Multi-Level Interpolation." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, Atlanta, GA, USA, Nov 17–22, 2024.
- C19 [ICS'24]** Jiajun Huang, Sheng Di, Xiaodong Yu, Yujia Zhai, Jinyang Liu, Yafan Huang, Ken Raffanetti, Hui Zhou, **Kai Zhao**, Xiaoyi Lu, Zizhong Chen, Franck Cappello, Yanfei Guo, Rajeev Thakur, "gZCCL: Compression-Accelerated Collective Communication Framework for GPU Clusters." *Proceedings of the ACM International Conference on Supercomputing*, Kyoto, Japan, June 4–7, 2024.
- C20 [IPDPS'24]** Zizhe Jian, Sheng Di, Jinyang Liu, **Kai Zhao**, Xin Liang, Haiying Xu, Robert Underwood, Shixun Wu, Jiajun Huang, Zizhong Chen, Franck Cappello, "CliZ: Optimizing Lossy Compression for Climate Datasets with Adaptive Fine-tuned Data Prediction." *Proceedings of the IEEE International Parallel and Distributed Processing Symposium*, San Francisco, CA, USA, May 27–31, 2024.
- C21 [IPDPS'24]** Jiajun Huang, Sheng Di, Xiaodong Yu, Yujia Zhai, Zhaorui Zhang, Jinyang Liu, Xiaoyi Lu, Ken Raffanetti, Hui Zhou, **Kai Zhao**, Zizhong Chen, Franck Cappello, Yanfei Guo, Rajeev Thakur, "An Optimized Error-controlled MPI Collective Framework Integrated with Lossy Compression." *Proceedings of the IEEE International Parallel and Distributed Processing Symposium*, San Francisco, CA, USA, May 27–31, 2024.
- C22 [ICDE'24]** Mingze Xia, Sheng Di, Franck Cappello, Pu Jiao, **Kai Zhao**, Jinyang Liu, Xuan Wu, Xin Liang, Hanqi Guo, "Preserving Topological Feature with Sign-of-Determinant Predicates in Lossy Compression: A Case Study of Vector Field Critical Points." *Proceedings of the IEEE International Conference on Data Engineering*, Utrecht, Netherlands, May 13–16, 2024.
- C23 [SIGMOD'24]** Jinyang Liu, Sheng Di, **Kai Zhao**, Xin Liang, Sian Jin, Zizhe Jian, Jiajun Huang, Shixun Wu, Zizhong Chen, Franck Cappello, "High-performance Effective Scientific Error-bounded Lossy Compression with Auto-tuned Multi-component Interpolation." *Proceedings of the ACM SIGMOD International Conference on Management of Data*, Santiago, Chile, June 9–15, 2024.
- C24 [HiPC'23]** Arham Khan, Sheng Di, **Kai Zhao**, Jinyang Liu, Kyle Chard, Ian Foster, Franck Cappello, "SECRE: Surrogate-based Error-controlled Lossy Compression Ratio Estimation Framework." *Proceedings of the IEEE International Conference on High Performance Computing, Data, and Analytics*, Goa, India, Dec 18–21, 2023.
- C25 [BigData'23]** Jiajun Huang, Jinyang Liu, Sheng Di, Yujia Zhai, Zizhe Jian, Shixun Wu, **Kai Zhao**, Zizhong Chen, Yanfei Guo, Franck Cappello, "Exploring Wavelet Transform Usages for Error-bounded Scientific Data Compression." *Proceedings of the IEEE International Conference on Big Data*, Sorrento, Italy, Dec 15–18, 2023.
- C26 [BigData'23]** Jinyang Liu, Sheng Di, Sian Jin, **Kai Zhao**, Xin Liang, Zizhong Chen, Franck Cappello, "Scientific Error-bounded Lossy Compression with Super-resolution Neural Networks." *Proceedings of the IEEE International Conference on Big Data*, Sorrento, Italy, Dec 15–18, 2023.
- C27 [SC'23]** Daoce Wang, Jesus Pulido, Pascal Grosset, Jiannan Tian, Sian Jin, Houjun Tang, Jean Sexton, Sheng

Di, **Kai Zhao**, Bo Fang, Zarija Lukić, Franck Cappello, James Ahrens, Dingwen Tao, “AMRIC: A Novel In Situ Lossy Compression Framework for Efficient I/O in Adaptive Mesh Refinement Applications.” *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, Denver, CO, USA, Nov 12–17, 2023.

- C28 [VLDB’23]** Pu Jiao, Sheng Di, Hanqi Guo, **Kai Zhao**, Jiannan Tian, Dingwen Tao, Xin Liang, Franck Cappello, “Toward Quantity-of-Interest Preserving Lossy Compression for Scientific Data.” *Proceedings of the International Conference on Very Large Data Bases*, Vancouver, Canada, Aug 28–Sep 1, 2023.
- C29 [ICS’23 (Best Paper Nominee)]** Jinyang Liu, Sheng Di, **Kai Zhao**, Xin Liang, Zizhong Chen, Franck Cappello, “FAZ: A Flexible Auto-Tuned Modular Error-Bounded Compression Framework for Scientific Data.” *Proceedings of the ACM International Conference on Supercomputing*, Orlando, FL, USA, June 21–23, 2023.
- C30 [CCGrid’23]** Yafan Huang, **Kai Zhao**, Sheng Di, Guanpeng Li, Maxim Dmitriev, Thierry-Laurent D. Tonellot, Franck Cappello, “Towards Improving Reverse Time Migration Performance by High-speed Lossy Compression.” *Proceedings of the IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing*, Bangalore, India, May 1–4, 2023.
- C31 [ICDE’23]** Md Hasanur Rahman, Sheng Di, **Kai Zhao**, Robert Underwood, Guanpeng Li, Franck Cappello, “A Feature-Driven Fixed-Ratio Lossy Compression Framework for Real-World Scientific Datasets.” *Proceedings of the IEEE International Conference on Data Engineering*, Anaheim, CA, USA, Apr 3–7, 2023.
- C32 [PPoPP’23]** Jieyang Chen, Xin Liang, **Kai Zhao**, Hadi Zamani Sabzi, Laxmi Bhuyan, Zizhong Chen, “Improving Energy Saving of One-sided Matrix Decompositions on CPU-GPU Heterogeneous Systems.” *Proceedings of the ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming*, Montreal, QC, Canada, Feb 25–Mar 1, 2023.
- C33 [HiPC’22]** Yuanjian Liu, Sheng Di, **Kai Zhao**, Sian Jin, Cheng Wang, Kyle Chard, Dingwen Tao, Ian Foster, Franck Cappello, “Optimizing Multi-Range based Error-Bounded Lossy Compression for Scientific Datasets.” *Proceedings of the IEEE International Conference on High Performance Computing, Data, and Analytics*, Bengaluru, India, Dec 17–20, 2022.
- C34 [SC’22]** Jinyang Liu, Sheng Di, **Kai Zhao**, Xin Liang, Zizhong Chen, Franck Cappello, “Dynamic Quality Metric Oriented Error Bounded Lossy Compression for Scientific Datasets.” *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, Dallas, TX, USA, Nov 13–18, 2022.
- C35 [HPDC’22]** Xiaodong Yu, Sheng Di, **Kai Zhao**, Jiannan Tian, Dingwen Tao, Xin Liang, Franck Cappello, “Ultra-fast Error-bounded Lossy Compression for Scientific Datasets.” *Proceedings of the ACM International Symposium on High-Performance Parallel and Distributed Computing*, Minneapolis, MN, USA, June 27–July 1, 2022.
- C36 [ICDE’22]** **Kai Zhao**, Sheng Di, Danny Perez, Xin Liang, Zizhong Chen, Franck Cappello, “MDZ: An Efficient Error-bounded Lossy Compressor for Molecular Dynamics.” *Proceedings of the IEEE International Conference on Data Engineering*, Kuala Lumpur, Malaysia, May 9–12, 2022.
- C37 [SC’21]** Sihuan Li, Sheng Di, **Kai Zhao**, Xin Liang, Zizhong Chen, Franck Cappello, “Resilient Error-bounded Lossy Compressor for Data Transfer.” *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, St. Louis, MO, USA, Nov 14–19, 2021.
- C38 [Cluster’21]** Jinyang Liu, Sheng Di, **Kai Zhao**, Sian Jin, Dingwen Tao, Xin Liang, Zizhong Chen, Franck Cappello, “Exploring Autoencoder-Based Error-Bounded Compression for Scientific Data.” *Proceedings of the IEEE International Conference on Cluster Computing*, Portland, OR, USA, Sep 7–10, 2021.
- C39 [Cluster’21]** Jiannan Tian, Sheng Di, Xiaodong Yu, Cody Rivera, **Kai Zhao**, Sian Jin, Yunhe Feng, Xin Liang, Dingwen Tao, Franck Cappello, “Optimizing Error-Bounded Lossy Compression for Scientific Data on GPUs.” *Proceedings of the IEEE International Conference on Cluster Computing*, Portland, OR, USA, Sep 7–10, 2021.
- C40 [ICS’21]** Yujia Zhai, Elisabeth Giem, Quan Fan, **Kai Zhao**, Jinyang Liu, Zizhong Chen, “FT-BLAS: A High Performance BLAS Implementation With Online Fault Tolerance.” *Proceedings of the ACM International Conference on Supercomputing*, Virtual Event, USA, June 14–17, 2021.
- C41 [ICDE’21]** **Kai Zhao**, Sheng Di, Maxim Dmitriev, Thierry-Laurent D. Tonellot, Zizhong Chen, Franck Cappello, “Optimizing Error-Bounded Lossy Compression for Scientific Data by Dynamic Spline Interpolation.” *Proceedings of the IEEE International Conference on Data Engineering*, Chania, Greece, Apr 19–22, 2021.
- C42 [PACT’20]** Jiannan Tian, Sheng Di, **Kai Zhao**, Cody Rivera, Megan Hickman, Robert Underwood, Sian Jin, Xin Liang, Jon Calhoun, Dingwen Tao, Franck Cappello, “cuSZ: An Efficient GPU-Based Error-Bounded Lossy Compression Framework for Scientific Data.” *Proceedings of the International Conference on Parallel Architectures and Compilation Techniques*, Virtual Event, GA, USA, Oct 3–7, 2020.
- C43 [Cluster’20]** Sihuan Li, Sheng Di, **Kai Zhao**, Xin Liang, Zizhong Chen, Franck Cappello, “Towards End-to-end

SDC Detection for HPC Applications Equipped with Lossy Compression.” *Proceedings of the IEEE International Conference on Cluster Computing*, Kobe, Japan, Sep 14–17, 2020.

- C44 [HPDC’20] Kai Zhao, Sheng Di, Xin Liang, Sihuan Li, Dingwen Tao, Zizhong Chen, Franck Cappello, “Significantly Improving Lossy Compression for HPC Datasets with Second-Order Prediction and Parameter Optimization.” *Proceedings of the ACM International Symposium on High-Performance Parallel and Distributed Computing*, Stockholm, Sweden, June 23–26, 2020.
- C45 [SC’19] Sihuan Li, Hongbo Li, Xin Liang, Jieyang Chen, Elisabeth Giem, Kaiming Ouyang, Kai Zhao, Sheng Di, Franck Cappello, Zizhong Chen, “FT-iSort: Efficient Fault Tolerance for Introsort.” *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, Denver, CO, USA, Nov 17–22, 2019.
- C46 [ICS’19] Jieyang Chen, Nan Xiong, Xin Liang, Dingwen Tao, Sihuan Li, Kaiming Ouyang, Kai Zhao, Nathan DeBardeleben, Qiang Guan, Zizhong Chen, “TSM2: Optimizing Tall-and-Skinny Matrix-Matrix Multiplication on GPUs.” *Proceedings of the ACM International Conference on Supercomputing*, Phoenix, AZ, USA, June 26–28, 2019.
- C47 [SC’18] Jieyang Chen, Hongbo Li, Sihuan Li, Xin Liang, Panruo Wu, Dingwen Tao, Kaiming Ouyang, Yuanlai Liu, Kai Zhao, Qiang Guan, Zizhong Chen, “Fault Tolerant One-sided Matrix Decompositions on Heterogeneous Systems with GPUs.” *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, Dallas, TX, USA, Nov 11–16, 2018.

REFEREED JOURNAL PUBLICATIONS

- J1 [TBD’26] Arham Khan, Sheng Di, Kai Zhao, Jinyang Liu, Kyle Chard, Ian Foster, Franck Cappello, “SLyCE: An Efficient Surrogate-based Lossy Compression Quality Estimation Framework.” *IEEE Transactions on Big Data*, 2026.
- J2 [CSUR’25] Sheng Di, Jinyang Liu, Kai Zhao, Xin Liang, Robert Underwood, Zhaorui Zhang, Milan Shah, Yafan Huang, Jiajun Huang, Xiaodong Yu, Congrong Ren, Hanqi Guo, Grant Wilkins, Dingwen Tao, Jiannan Tian, Sian Jin, Zizhe Jian, Daoce Wang, Md Hasanur Rahman, Boyuan Zhang, Shihui Song, Jon C. Calhoun, Guanpeng Li, Kazutomo Yoshii, Khalid Ayed Alharthi, Franck Cappello, “A Survey on Error-Bounded Lossy Compression for Scientific Datasets.” *ACM Computing Surveys*, 2025.
- J3 [TPDS’25] Zhaorui Zhang, Sheng Di, Kai Zhao, Sian Jin, Dingwen Tao, Zhuoran Ji, Benben Liu, Khalid Alharthi, Jiannong Cao, Franck Cappello, “FedCSpc: A Cross-Silo Federated Learning System with Error-Bounded Lossy Parameter Compression.” *IEEE Transactions on Parallel and Distributed Systems*, 2025.
- J4 [TPDS’24] Daoce Wang, Jesus Pulido, Pascal Grosset, Sian Jin, Jiannan Tian, Kai Zhao, James Ahrens, Dingwen Tao, “TAC+: Optimizing Error-Bounded Lossy Compression for 3D AMR Simulations.” *IEEE Transactions on Parallel and Distributed Systems*, 2024.
- J5 [TPDS’23] Yujia Zhai, Elisabeth Giem, Kai Zhao, Jinyang Liu, Jiajun Huang, Bryan Wong, Christian Shelton, Zizhong Chen, “FT-BLAS: A Fault Tolerant High Performance BLAS Implementation on x86 CPUs.” *IEEE Transactions on Parallel and Distributed Systems*, 2023.
- J6 [TBD’22 (Best Paper Award)] Xin Liang*, Kai Zhao*, Sheng Di, Sihuan Li, Robert Underwood, Ali Murat Gok, Jiannan Tian, Junjing Deng, Jon C. Calhoun, Dingwen Tao, Zizhong Chen, Franck Cappello, “SZ3: A Modular Framework for Composing Prediction-Based Error-Bounded Lossy Compressors.” *IEEE Transactions on Big Data*, 2022. *Authors contributed equally.
- J7 [TPDS’22] Yuanjian Liu, Sheng Di, Kai Zhao, Sian Jin, Cheng Wang, Kyle Chard, Dingwen Tao, Ian Foster, Franck Cappello, “Optimizing Error-Bounded Lossy Compression for Scientific Data With Diverse Constraints.” *IEEE Transactions on Parallel and Distributed Systems*, 2022.
- J8 [TPDS’21] Kai Zhao, Sheng Di, Sihuan Li, Xin Liang, Yujia Zhai, Jieyang Chen, Kaiming Ouyang, Franck Cappello, Zizhong Chen, “Algorithm-Based Fault Tolerance for Convolutional Neural Networks.” *IEEE Transactions on Parallel and Distributed Systems*, 2021.

REFEREED WORKSHOP PUBLICATIONS

- W1 [DRBSD-11] Ziwei Qiu, Jinyang Liu, Kai Zhao, Robert Underwood, Sheng Di, “Benchmarking Cutting-Edge Scientific Error-Bounded Lossy Compressors on Correlation-Based Rate-Distortion.” *Proceedings of the International Workshop on Data Analysis and Reduction for Big Scientific Data @SC’25*, St. Louis, MO, USA, Nov 17, 2025.
- W2 [DRBSD-10] Tripti Agarwal, Sheng Di, Jiajun Huang, Yafan Huang, Ganesh Gopalakrishnan, Robert Under-

wood, **Kai Zhao**, Xin Liang, Guanpeng Li, Franck Cappello, “SZOps: Scalar Operations for Error-bounded Lossy Compressor for Scientific Data.” *Proceedings of the International Workshop on Data Analysis and Reduction for Big Scientific Data @SC’24*, Atlanta, GA, USA, Nov 18, 2024.

- W3 [Cluster-W’23]** Arham Khan, Sheng Di, **Kai Zhao**, Jinyang Liu, Kyle Chard, Ian Foster, Franck Cappello, “An Efficient and Accurate Compression Ratio Estimation Model for SZx.” *Proceedings of the IEEE International Conference on Cluster Computing Workshops*, Santa Fe, NM, USA, Oct 31–Nov 3, 2023.
- W4 [IWBD-2]** Jinyang Liu, Sihuan Li, Sheng Di, Xin Liang, **Kai Zhao**, Dingwen Tao, Zizhong Chen, Franck Cappello, “Improving Lossy Compression for SZ by Exploring the Best-Fit Lossless Compression Techniques.” *Proceedings of the International Workshop on Big Data Reduction @BigData’21*, Orlando, FL, USA, Dec 15–18, 2021.
- W5 [IWBD-1]** **Kai Zhao**, Sheng Di, Xin Liang, Sihuan Li, Dingwen Tao, Julie Bessac, Zizhong Chen, Franck Cappello, “SDRBench: Scientific Data Reduction Benchmark for Lossy Compressors.” *Proceedings of the International Workshop on Big Data Reduction @BigData’20*, Atlanta, GA, USA, Dec 10–13, 2020.
- W6 [DRBSD-7]** Yuanjian Liu, Sheng Di, **Kai Zhao**, Sian Jin, Cheng Wang, Kyle Chard, Dingwen Tao, Ian Foster, Franck Cappello, “Understanding Effectiveness of Multi-error-bounded Lossy Compression for Preserving Ranges of Interest in Scientific Analysis.” *Proceedings of the International Workshop on Data Analysis and Reduction for Big Scientific Data @SC’21*, St. Louis, MO, USA, Nov 14, 2021.

MENTORING

- **Ph.D.:** Longtao Zhang (2023–present), Ruoyu Li (2023–present), Zhuoxun Yang (2024–present), Chuan Yu (2026–present), Md Mehrab Hossain Opi (2026–present)
- **M.S.:** Nicholas Simmons (2024)
- **Student honors:** FSU Dean’s Award for Doctoral Excellence and Naaman Franklin Faile Jr. Graduate Fellowship (Zhang, 2025; Li, 2026); FSU Graduate School Legacy Fellowship (Opi, 2026); FSU Computer Science Research Assistant Award (Zhang, 2025); Argonne Training Program on Extreme-Scale Computing (Zhang, Li, 2026); Argonne National Laboratory research internships (Zhang, Li, 2025–present); SC’24 Travel Grant (Li)
- **Ph.D. committee member:** Tusher Chandra Mondol (FSU)

TEACHING

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|--|----------------------|
| ■ FSU COP 4521 Programming Secure, Parallel, and Distributed Applications | Spring 2026 |
| ■ FSU ISC 5318 High Performance Computing | Spring 2025 |
| ■ FSU CIS 4930/5930 Special Topics in Computer Science: High Performance Computing | Spring 2024 |
| ■ FSU CDA 4150/5155 Computer Architecture | Fall 2023, Fall 2025 |
| ■ UAB CS 632/732 Parallel Computing | Spring 2023 |
| ■ UAB CS 332/532 Systems Programming | Fall 2022 |

SERVICE

- **Organizing Committee:** HPC Asia’27 (track co-chair, Hardware Systems & Architectures), IWBD-23 (co-chair), NSF FZ Workshop’24 and FZ/ZF Joint Workshop’25 (host, Sarasota, FL)
- **Program Committee:** HPCC, IWBD-23
- **Reviewer:** SC, ICS, HPDC, IPDPS, ICPP, CCGrid, BigData, ICMLA, ICSNC, SELSE; TPDS, TC, TBD, TACO, ParCo, TPAMI, UAAI
- **FSU:** Financial Aid & Admissions Committee (2023–present), Graduate Studies Committee (2023–present), Faculty Recruitment Committee (2025–present), CS Expo’25 (co-chair)
- **UAB:** Department Seminar Coordinator, Research Infrastructure Coordinator (2022–2023)